Cut the Line · Defense Sheet

Real-time protection against phone scams and AI-cloned voices

Hackathon PoC
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THREAT SCENARIO

A scammer calls an elderly person posing as a grandchild, a bank advisor or the police, sometimes with an **AI-cloned voice** or a **spoofed number**, and pushes for an urgent, secret payment. Cut the Line sits on the call path: it analyzes unknown callers live, warns the callee, puts the caller on hold, and verifies identity with a 4-digit code texted to the **real** relative's phone. A voice clone cannot pass it.

NIST CSF 2.0 FUNCTIONS

Function	What Cut the Line does
Govern	Caller notified before any analysis; no audio or transcript kept; metadata purged after 30 days
Identify	Trusted contact book; number reputation (VoIP, hidden, identity vs number)
Protect	Out-of-band code to the real relative; PIN-locked app; signed Twilio webhooks
Detect	4 live signals: Claude content 45, keywords 20, synthetic voice 20, reputation 15
Respond	Score 40 caution · 65 warning to callee + text to relative · 85 caller on hold. Never auto hang-up
Recover (partial)	Resume after verification; history with risk peaks and actions

ATT&CK · ATLAS · OWASP

Technique / risk	Mitigation
T1566.004 · T1598.004 Spearphishing Voice	Live content analysis; requests for codes or bank details trip a circuit breaker
T1684.001 Impersonation	Claimed identity checked against the contact book; code sent to the real relative
ATLAS AML.T0052.001 · T0088 Deepfake phishing	Voice signal capped at 64 alone; identity proven by the real phone, not the voice
OWASP LLM01 Prompt injection	Transcript treated as untrusted data; LLM is one signal of four
LLM05 · LLM06	JSON-only, clamped output; a fixed policy, not the LLM, triggers actions
LLM02	No audio, transcript or code stored

METRICS

22 scripted calls (11 scam / 11 legit)	Detected	False pos.
Keywords only	0 / 11	0 / 11
Offline stand-in for Claude	3 / 11	0 / 11
Claude on Bedrock	to run	to run

Time to detect: 15 s into the grandparent scam (score 85); 64 s for the fake bank advisor.
Time to respond: hold + warning 0.3 s after the threshold (replay). Real network latency not yet measured.

1,024 automated tests pass. Scenarios are synthetic: they validate the pipeline, not field accuracy.

KNOWN LIMITATIONS

- **One US relay number**; a carrier could run the same logic natively.
- **Single user, one call at a time**; no database yet.
- **Web app rings only while open** (no VoIP push without a native app).
- **Spoofed trusted numbers** are not analyzed; the code check is the defense.
- **Voice model** needs calibration on real phone audio.
- **Amazon Transcribe** reuses audio unless an AWS AI opt-out policy is applied.